Lab 9: Estimating and Stress-Testing an RD

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Wenhao Jiang

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1 Goals

1. Draw the picture first, because in an RD the picture *is* the result.
2. Estimate with local linear regression and robust bias-corrected intervals.
3. Choose a bandwidth and a polynomial order, and see how much the answer depends on them.
4. Run every validity check an RD paper should report, including covariate balance.
5. Add pre-treatment covariates for precision.
6. See in simulation what manipulation of the running variable does.
7. Estimate a fuzzy RD on real data, step by step: first stage, reduced form, ratio, local 2SLS, checks.

```
library(dplyr)
library(ggplot2)
library(rdrobust)
library(rddensity)

data("rdrobust_RDsenate", package = "rdrobust")
sen <- rdrobust_RDsenate |> filter(!is.na(vote))
c(n = nrow(sen), min_margin = min(sen$margin), max_margin = max(sen$margin))
#>          n min_margin max_margin
#>       1297       -100        100
```

2 The design

Question. Does winning a Senate seat make a party more likely to win that seat again six years later — the *incumbency advantage*?

Running variable. `margin`: the Democratic vote-share margin of victory in the previous election. **Cutoff.** Zero. **Treatment.** Democratic incumbency. **Outcome.** `vote`: the Democratic vote share in the following election.

The identification claim is that elections decided by a fraction of a percentage point are essentially coin flips, so parties that barely won and parties that barely lost are comparable in everything except incumbency.

3 The picture comes first

```
rdplot(y = sen$vote, x = sen$margin, c = 0,
       x.label = "Democratic margin of victory, previous election",
       y.label = "Democratic vote share, next election",
       title = "")
```

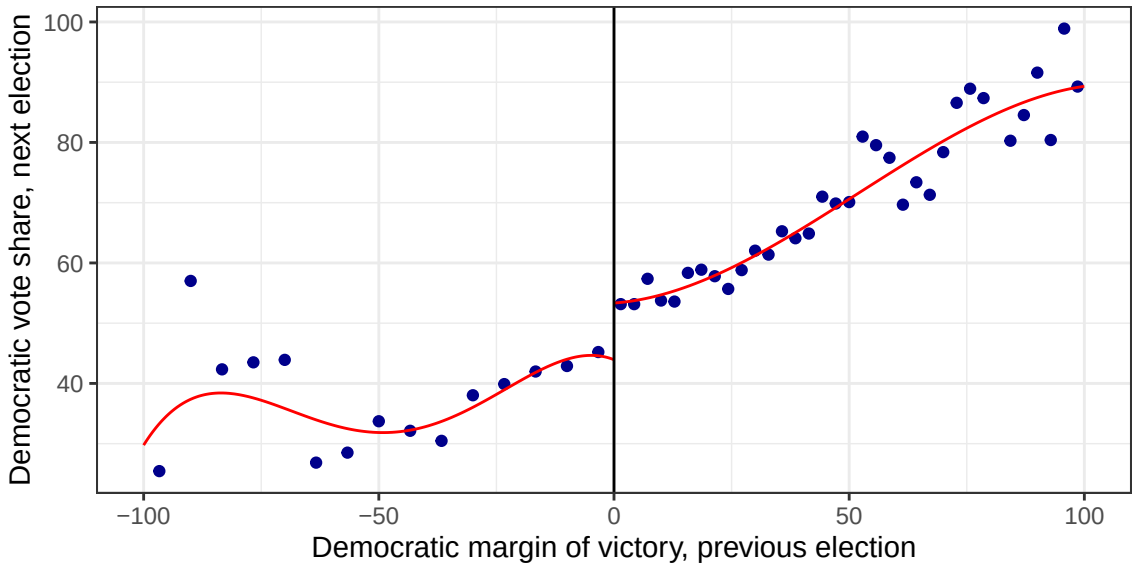


Figure 1: Binned means of the following election's Democratic vote share against the previous margin of victory, with fourth-order polynomials fitted separately on each side.

> **!** Important
>
> Look at the jump at zero before you compute anything. If you cannot see it here, no amount of bandwidth tuning should convince you it is there. Conversely, a jump this visible does not need a high-order polynomial to find it.

```
rdplot(y = sen$vote[abs(sen$margin) < 15], x = sen$margin[abs(sen$margin) <
↪  15],
       c = 0, p = 1, nbins = c(20, 20),
       x.label = "margin", y.label = "next vote share", title = "")
```

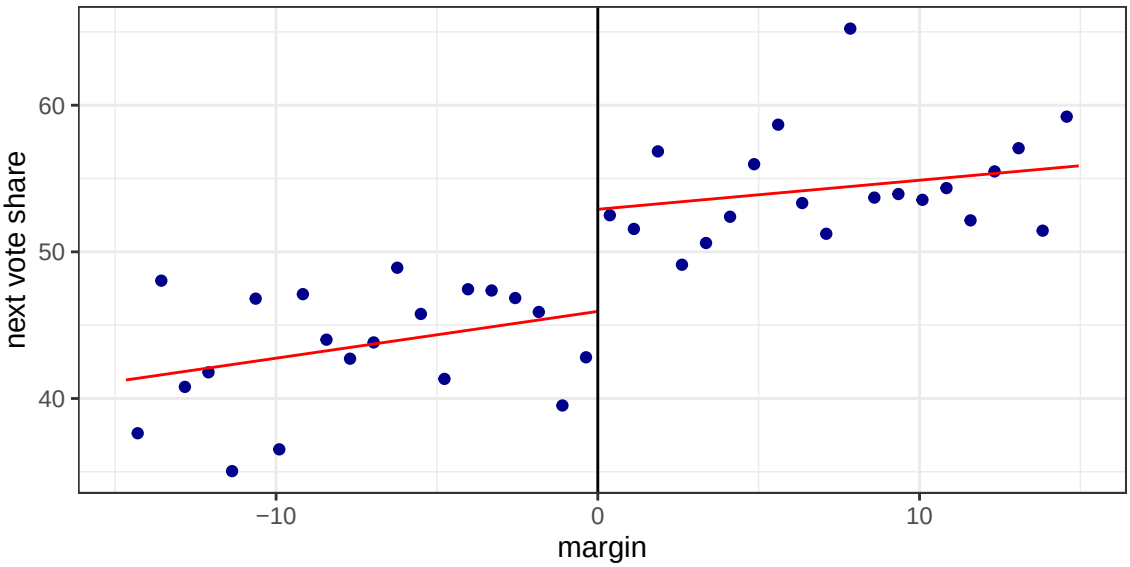


Figure 2: The same data with local linear fits inside a 15-point window. This is closer to what the estimator actually uses.

4 Estimation

```
rd <- rdrobust(y = sen$vote, x = sen$margin, c = 0)
summary(rd)
#> Call: rdrobust
#>
#> Sharp RD estimates using local polynomial regression.
#>
#> Number of Obs.                 1297
#> BW type                       mserd
#> Kernel                   Triangular
#> VCE method                       NN
#>
#>                                Left        Right
#> Number of Obs.                  595          702
#> Eff. Number of Obs.             360          323
#> Order est. (p)                    1            1
#> Order bias (q)                    2            2
#> BW est. (h)                  17.754       17.754
#> BW bias (b)                  28.028       28.028
#> rho (h/b)                     0.633        0.633
#> Unique Obs.                     595          665
#>
#> ===============================================================================
```

```
#>                     Point    Robust Inference
#>                  Estimate         z     P>|z|      [ 95% C.I. ]
#> ---------------------------------------------------------------------
#>      RD Effect     7.414     4.311     0.000     [4.094 , 10.919]
#> =====================================================================
```

Three numbers to read from that output.

- **Conventional**: the local linear estimate at the MSE-optimal bandwidth, with a naive standard error.
- **Bias-Corrected**: the point estimate after subtracting an estimate of the leading bias term.
- **Robust**: the confidence interval that accounts for the *uncertainty in that bias correction.* **This is the one to report** (Calonico, Cattaneo, and Titiunik 2014).

```
## how many observations are actually doing the work?
c(bandwidth_left = rd$bws[1, 1], bandwidth_right = rd$bws[1, 2],
  n_left = rd$N_h[1], n_right = rd$N_h[2], n_total = sum(rd$N))
#>  bandwidth_left bandwidth_right          n_left         n_right
#>           17.75           17.75          360.00          323.00
#>         n_total
#>         1297.00
```

Note the gap between the nominal sample size and the effective one. An RD with 1,300 observations may be estimated from 400.

4.1 When the conventional and robust rows disagree

```
## the three rows side by side: estimate, s.e., p-value, 95% interval
round(cbind(estimate = rd$coef, se = rd$se, p = rd$pv, rd$ci), 3)
#>                Coeff Std. Err. P>|z| CI Lower CI Upper
#> Conventional   7.414     1.459     0    4.555    10.27
#> Bias-Corrected 7.507     1.459     0    4.647    10.37
#> Robust         7.507     1.741     0    4.094    10.92
```

The robust row changes two things. Its centre is the bias-corrected estimate, and its standard error is larger, because it also counts the noise in the estimated bias. In the Senate data every row is far from zero. In smaller or noisier applications they can split: the conventional interval excludes zero and the robust one does not. Report the robust interval. The conventional one ignores the bias at the MSE-optimal bandwidth and covers less than 95% (Week 9).

4.2 Doing it by hand

`rdrobust` is a black box until you have written the estimator once.

```r
h <- rd$bws[1, 1]                       # use the same bandwidth

sub <- sen |> filter(abs(margin) <= h) |>
  mutate(D = as.integer(margin >= 0),
         kw = 1 - abs(margin) / h)      # triangular kernel

## fully interacted local linear regression: separate slopes each side
fit_manual <- lm(vote ~ D * margin, data = sub, weights = kw)

c(manual = coef(fit_manual)["D"], rdrobust_conventional = rd$coef[1])
#>              manual.D rdrobust_conventional
#>                 7.414                 7.414
```

The coefficient on the treatment indicator in a kernel-weighted, fully interacted local linear regression *is* the RD estimate. Everything `rdrobust` adds is bandwidth selection and bias correction.

4.3 Bandwidth sensitivity

```r
bws <- seq(3, 40, by = 1)
sens <- lapply(bws, function(b) {
  r <- rdrobust(y = sen$vote, x = sen$margin, c = 0, h = b)
  tibble::tibble(h = b, est = r$coef[3],          # bias-corrected point
↪  estimate
                 lo = r$ci[3, 1], hi = r$ci[3, 2]) # matching robust interval
}) |> bind_rows()

ggplot(sens, aes(h, est)) +
  geom_ribbon(aes(ymin = lo, ymax = hi), fill = "navy", alpha = .15) +
  geom_line(colour = "navy", linewidth = .8) +
  geom_hline(yintercept = 0, linetype = 2) +
  geom_vline(xintercept = rd$bws[1, 1], colour = "firebrick", linetype = 3) +
  labs(x = "bandwidth", y = "RD estimate") +
  theme_minimal(base_size = 10)
```

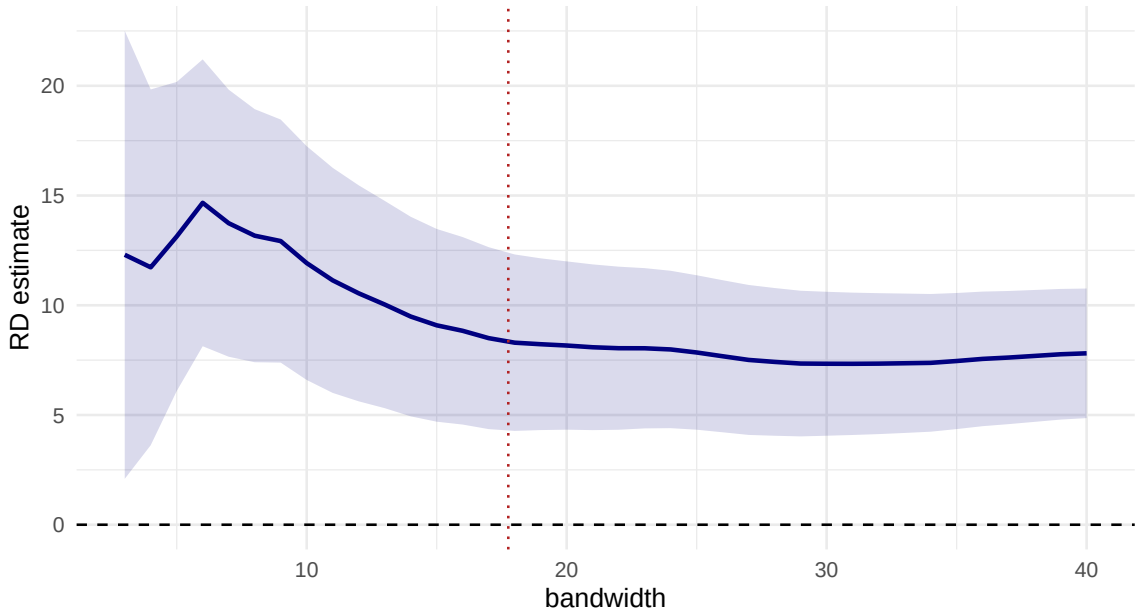


Figure 3: The RD estimate across bandwidths from 3 to 40 points, with robust 95% intervals. A credible result is flat over a wide range.

The dotted red line marks the MSE-optimal bandwidth. The estimate is stable across a wide range — this is what a robust RD looks like.

4.4 A coverage-optimal bandwidth

The default bandwidth minimizes the MSE of the estimate. For inference, `bwselect = "cerrd"` picks a smaller bandwidth that makes the robust interval's coverage error smallest (h_{CER}, Week 9).

```
## same estimator, coverage-optimal bandwidth
rd_cer <- rdrobust(y = sen$vote, x = sen$margin, c = 0, bwselect = "cerrd")
## compare bandwidths, bias-corrected estimates, and robust intervals
round(rbind(mse = c(h = rd$bws[1, 1], estimate = rd$coef[2], rd$ci[3, ]),
            cer = c(rd_cer$bws[1, 1], rd_cer$coef[2], rd_cer$ci[3, ])), 2)
#>         h estimate CI Lower CI Upper
#> mse 17.75     7.51     4.09    10.92
#> cer 12.41     7.68     4.07    11.29
```

A smaller window means fewer races, so the interval widens; the estimate should not move much. Reporting both is common.

4.5 Local polynomial order

Local linear (`p = 1`) is the default. A local quadratic is the usual check.

```
## re-estimate with p = 1 and p = 2; each gets its own MSE-optimal bandwidth
ord <- sapply(1:2, function(pp) {
  r <- rdrobust(y = sen$vote, x = sen$margin, c = 0, p = pp)
  c(p = pp, estimate = r$coef[1], h = r$bws[1, 1],
    robust_lo = r$ci[3, 1], robust_hi = r$ci[3, 2])
})
round(t(ord), 2)
#>      p estimate     h robust_lo robust_hi
#> [1,] 1     7.41 17.75      4.09     10.92
#> [2,] 2     8.05 22.26      4.21     12.42
```

The quadratic chooses a wider bandwidth, because a curve can follow the data farther from the cutoff, but its interval is wider: the extra curvature costs variance at the edge. A stable estimate across $p = 1, 2$ is what we want. Orders above two are rarely used, for the reason in the next section.

4.6 Why not a global polynomial?

```
poly_est <- sapply(1:8, function(p) {
  m <- lm(vote ~ poly(margin, p, raw = TRUE) * I(margin >= 0), data = sen)
  coef(m)["I(margin >= 0)TRUE"]
})
round(setNames(poly_est, paste0("degree_", 1:8)), 3)
#> degree_1 degree_2 degree_3 degree_4 degree_5 degree_6 degree_7 degree_8
#>    6.044    4.935    7.319    9.407    7.958    6.627    8.930    9.099
```

The answer wanders with the polynomial degree, and every one of these estimates uses observations 80 points from the cutoff to fit the value *at* the cutoff. Gelman and Imbens (2019) explain why the implied weights are indefensible.

5 Validity checks

5.1 Is the density of the running variable continuous?

If parties could precisely control their vote margin, we would see bunching just above zero.

```
dens <- rddensity(X = sen$margin, c = 0)
summary(dens)
#>
#> Manipulation testing using local polynomial density estimation.
#>
#> Number of obs =       1297
#> Model =               unrestricted
```

```
#> Kernel =              triangular
#> BW method =           estimated
#> VCE method =          jackknife
#>
#> c = 0                 Left of c           Right of c
#> Number of obs         595                 702
#> Eff. Number of obs    375                 376
#> Order est. (p)        2                   2
#> Order bias (q)        3                   3
#> BW est. (h)           18.439              22.774
#>
#> Method                T                   P > |T|
#> Robust                -0.76               0.4473
#>
#> P-values of binomial tests (H0: p=0.5).
#>
#> Window Length / 2          <c     >=c    P>|T|
#> 1.080                       20      28    0.3123
#> 2.160                       49      50    1.0000
#> 3.240                       85      73    0.3816
#> 4.320                      113     100    0.4110
#> 5.400                      136     123    0.4559
#> 6.480                      154     140    0.4484
#> 7.560                      188     159    0.1327
#> 8.640                      220     178    0.0397
#> 9.720                      240     201    0.0702
#> 10.800                     257     224    0.1445
```

```
rdplotdensity(dens, sen$margin, plotN = 100)
#> $Estl
#> Call: lpdensity
#>
#> Sample size                                    595
#> Polynomial order for point estimation    (p=)  2
#> Order of derivative estimated            (v=)  1
#> Polynomial order for confidence interval (q=)  3
#> Kernel function                                triangular
#> Scaling factor                                 0.459104938271605
#> Bandwidth method                               user provided
#>
#> Use summary(...) to show estimates.
#>
```

```
#> $Estr
#> Call: lpdensity
#>
#> Sample size                                        702
#> Polynomial order for point estimation    (p=)      2
#> Order of derivative estimated            (v=)      1
#> Polynomial order for confidence interval (q=)      3
#> Kernel function                                    triangular
#> Scaling factor                                     0.541666666666667
#> Bandwidth method                                   user provided
#>
#> Use summary(...) to show estimates.
#>
#> $Estplot
```

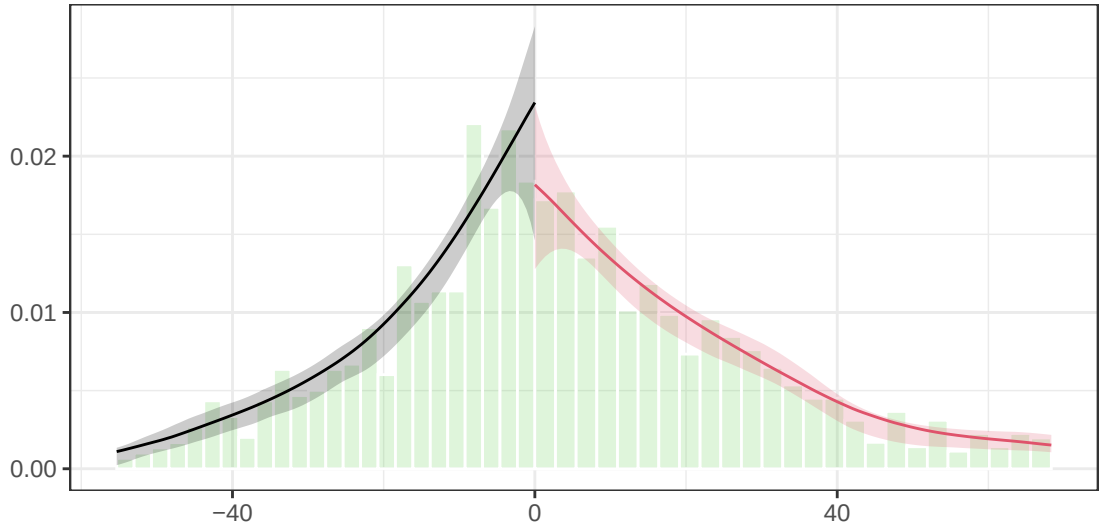


Figure 4: Estimated density of the running variable on each side of the cutoff with confidence bands. Overlapping bands at zero are what we want to see.

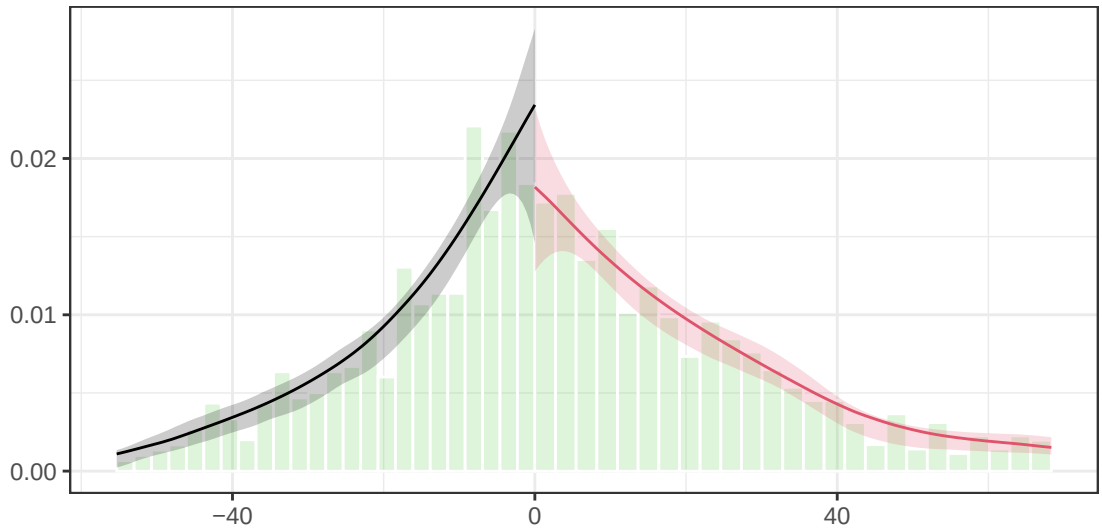


Figure 5: Estimated density of the running variable on each side of the cutoff with confidence bands. Overlapping bands at zero are what we want to see.

Non-rejection is consistent with the design, but it is not a validation of it: the test has limited power, and it detects only manipulation that shows up as bunching in the density. In settings where actors *can* manipulate the running variable (income thresholds, test retakes) treat it as one weak signal among several.

5.2 Covariate balance

Variables fixed before the election cannot be affected by who won it, so they should not jump at zero. Run the RD with each one as the outcome.

```
## pre-election covariates: Democratic vote share 6 and 12 years earlier,
## the presidential vote, and whether the seat was open
cv <- c("demvoteshlag1", "demvoteshlag2", "presdemvoteshlag1", "dopen")
bal <- sapply(cv, function(z) {
  ok <- !is.na(sen[[z]])                          ## drop missing values first
  r <- rdrobust(y = sen[[z]][ok], x = sen$margin[ok], c = 0,
                bwselect = "cerrd")
  ## bias-corrected jump, robust interval and p-value
  c(estimate = r$coef[2], robust_lo = r$ci[3, 1], robust_hi = r$ci[3, 2],
    p_robust = r$pv[3])
})
round(t(bal), 3)
#>                   estimate robust_lo robust_hi p_robust
#> demvoteshlag1        3.717    -1.710     9.145    0.179
#> demvoteshlag2        2.148    -2.326     6.622    0.347
#> presdemvoteshlag1   -0.088    -3.791     3.615    0.963
#> dopen               -0.139    -0.301     0.023    0.093
```

No covariate jumps at the 5% level. With several covariates, expect about one in twenty to "fail" by chance; a large, clearly significant jump in a pre-election variable would be the warning sign.

5.3 Placebo cutoffs

The estimator should find nothing at thresholds where nothing happens.

```
placebos <- c(-20, -15, -10, -5, 5, 10, 15, 20)
plc <- sapply(placebos, function(cc) {
  d <- if (cc < 0) sen |> filter(margin < 0) else sen |> filter(margin >= 0)
  r <- rdrobust(y = d$vote, x = d$margin, c = cc)
  c(estimate = r$coef[1], p_robust = r$pv[3])
})
round(t(plc), 3) |> `rownames<-`(paste("cutoff", placebos))
#>            estimate p_robust
#> cutoff -20    6.620    0.164
#> cutoff -15   -6.893    0.233
#> cutoff -10  -23.216    0.005
#> cutoff -5    -0.177    0.722
#> cutoff 5      3.321    0.673
#> cutoff 10     2.915    0.108
#> cutoff 15    -1.218    0.648
#> cutoff 20    -0.250    0.894
```

Note the restriction to one side of the true cutoff: otherwise a placebo test would pick up the genuine discontinuity at zero.

5.4 Donut hole

Drop the observations nearest the cutoff, where manipulation would be concentrated.

```
donut <- sapply(c(0, 0.25, 0.5, 1), function(dd) {
  d <- sen |> filter(abs(margin) > dd)
  r <- rdrobust(y = d$vote, x = d$margin, c = 0)
  c(estimate = r$coef[1], ci_lo = r$ci[3, 1], ci_hi = r$ci[3, 2])
})
round(t(donut), 3) |> `rownames<-`(paste("hole =", c(0, 0.25, 0.5, 1)))
#>             estimate ci_lo ci_hi
#> hole = 0       7.414 4.094 10.92
#> hole = 0.25    6.936 3.315 10.45
#> hole = 0.5     7.092 3.196 10.83
#> hole = 1       6.581 2.187 10.60
```

6 Adding covariates

Pre-treatment covariates can soak up variation in the outcome and tighten the interval. Because they do not jump at the cutoff, they should barely move the estimate. Use the same races with and without them, so the comparison is fair.

```
## keep races with all four covariates observed
sc <- sen[complete.cases(sen[, cv]), ]
## without and with covariates, both at the coverage-optimal bandwidth
r0 <- rdrobust(y = sc$vote, x = sc$margin, c = 0, bwselect = "cerrd")
rz <- rdrobust(y = sc$vote, x = sc$margin, c = 0, bwselect = "cerrd",
               covs = as.matrix(sc[, cv]))
round(rbind(no_covariates = c(estimate = r0$coef[2], robust_se = r0$se[3],
                              r0$ci[3, ]),
            covariates = c(rz$coef[2], rz$se[3], rz$ci[3, ])), 2)
#>                estimate robust_se CI Lower CI Upper
#> no_covariates      8.19      1.90     4.47    11.90
#> covariates         7.45      1.86     3.81    11.09
```

Only pre-treatment variables belong in `covs`. A variable the election could affect (turnout, spending after the election) would be a bad control. If the estimate moves a lot when covariates enter, suspect that one of them jumps.

7 What manipulation actually does

Simulate a world where the treatment has **no effect at all**, but where units just below the cutoff can push themselves above it.

```
set.seed(723)
n <- 4000
u <- rnorm(n)                                # ability: raises both X and Y
x_true <- rnorm(n) + 0.8 * u

## units with high u who are just below the cutoff push themselves over
push <- (x_true > -0.3 & x_true < 0) & (u > 0.5)
x <- ifelse(push, abs(x_true) + 0.01, x_true)

d <- as.integer(x >= 0)
y <- 0 * d + u + rnorm(n)                    # TRUE EFFECT IS ZERO

r_bad <- rdrobust(y = y, x = x, c = 0)
c(estimate = r_bad$coef[1], p_robust = r_bad$pv[3], truth = 0)
#>  estimate  p_robust     truth
#> 0.4515959 0.0007735 0.0000000
```

```
dt <- rddensity(X = x, c = 0)
c(t_statistic = dt$test$t_jk, p_value = dt$test$p_jk)
#> t_statistic     p_value
#>     2.19083     0.02846
```

The RD finds a large "effect" that does not exist — and **the density test catches it**. That is why the test belongs in every RD paper — while remembering that manipulation which does *not* produce bunching would slip past it.

```
tibble::tibble(x) |> filter(abs(x) < 1.5) |>
  ggplot(aes(x)) +
  geom_histogram(bins = 60, fill = "navy", alpha = .7) +
  geom_vline(xintercept = 0, colour = "firebrick", linewidth = .8) +
  labs(x = "running variable", y = "count") +
  theme_minimal(base_size = 10)
```

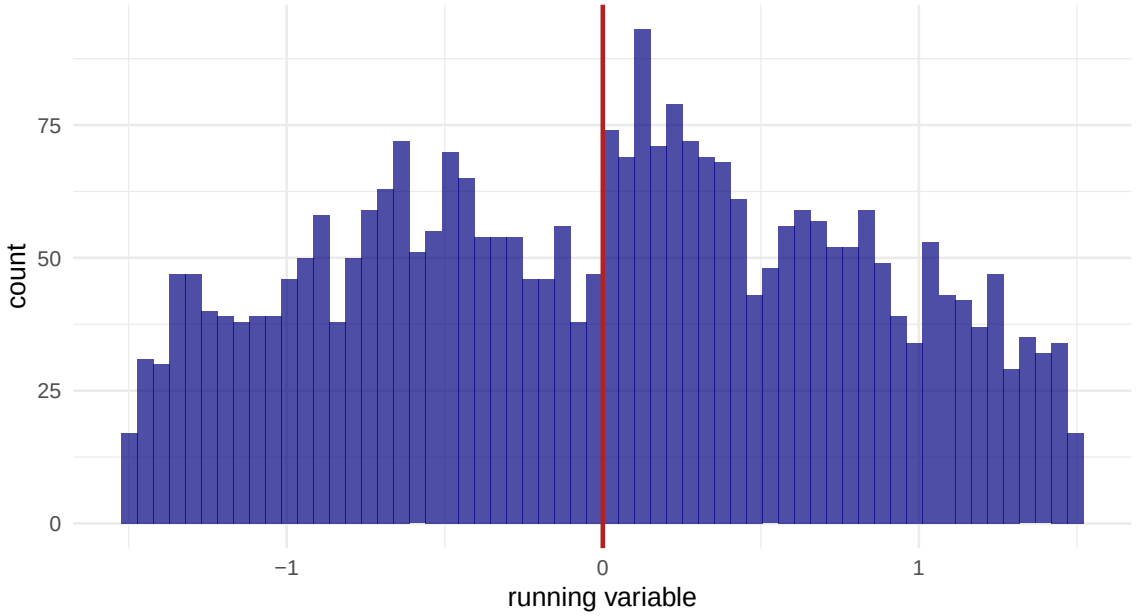


Figure 6: Histogram of the running variable under manipulation. The excess mass just above zero is the signature.

8 Fuzzy RD

When crossing the cutoff changes the *probability* of treatment, `rdrobust` takes a `fuzzy` argument and returns the ratio estimate.

8.1 A simulation

```
set.seed(1)
n <- 3000
xx <- runif(n, -1, 1)
takeup <- rbinom(n, 1, ifelse(xx >= 0, 0.75, 0.20))   # jump of 0.55
yy <- 1.5 * takeup + 0.5 * xx + rnorm(n)               # TRUE effect = 1.5

## sharp (intention-to-treat): attenuated toward zero
r_itt   <- rdrobust(y = yy, x = xx, c = 0)
## fuzzy: scaled up by the first stage
r_fuzzy <- rdrobust(y = yy, x = xx, c = 0, fuzzy = takeup)

c(ITT = r_itt$coef[1], fuzzy = r_fuzzy$coef[1], truth = 1.5)
#>    ITT  fuzzy  truth
#> 0.8771 1.4247 1.5000
```

The sharp estimate is the *intention-to-treat* effect of crossing the cutoff. The fuzzy estimate divides it by the jump in take-up, exactly as in Week 7 — and it inherits the LATE interpretation. Here the estimand is the effect for compliers *at the cutoff*.

8.2 A real application: financial aid in Colombia

Londoño-Vélez, Rodríguez, and Sánchez (2020, *AEJ: Economic Policy*) study *Ser Pilo Paga* (SPP), a Colombian program that paid full tuition at a high-quality university for students who were both high-achieving and poor. Two cutoffs decide eligibility: a national exam score (top 9%) and a household wealth index, SISBEN, which must fall below a regional threshold. We use the replication file from Cattaneo, Idrobo, and Titiunik (2024, *A Practical Introduction to RD Designs: Extensions*), which keeps only students above the exam cutoff, so SISBEN is the single running variable.

variable	meaning
`X1`	distance of the household's SISBEN index from its regional cutoff; positive = eligible
`T`	eligible (1 if `X1` ≥ 0): the instrument
`D`	actually received SPP: the treatment
`Y`	enrolled in a higher-education institution right after the exam
`icfes_*`	pre-determined: female, age, ethnic minority, residential stratum, family size

The design is fuzzy because eligibility is not receipt: an eligible student still had to be admitted to a certified university and enroll there.

```r
spp <- read.csv("../Data/spp_colombia.csv")
nrow(spp)
#> [1] 23132
## who received the subsidy, by eligibility
with(spp, table(eligible = T, received = D))
#>          received
#> eligible    0    1
#>        0 7709    0
#>        1 6258 9165
```

No ineligible student received SPP: non-compliance is **one-sided**. There are no always-takers and no defiers, so monotonicity holds by construction, and the compliers are all the students who received the subsidy near the cutoff.

8.2.1 Step 1: the first stage

```r
## first stage in a picture: binned means of D
rdplot(y = spp$D, x = spp$X1, c = 0,
       x.label = "SISBEN distance to cutoff (positive = eligible)",
       y.label = "received SPP", title = "")
```

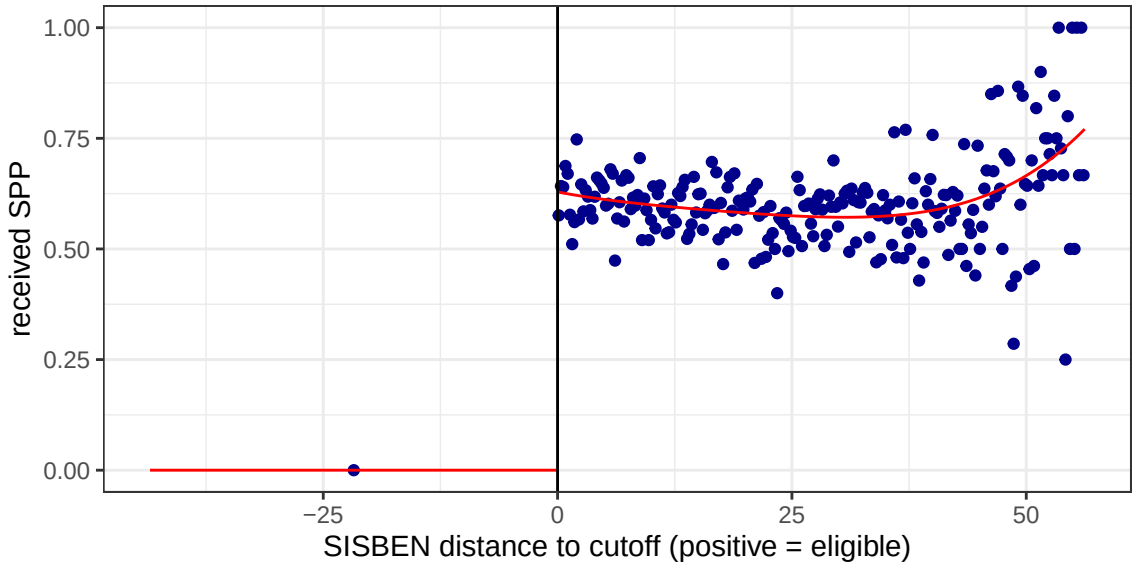


Figure 7: Share receiving SPP against distance to the SISBEN cutoff. Zero on the left by rule; a jump of about 0.6 at the cutoff.

```r
## first stage as a sharp RD with D as the outcome
fs <- rdrobust(y = spp$D, x = spp$X1, c = 0)
round(c(jump_in_D = fs$coef[1], robust_lo = fs$ci[3, 1],
```

```
        robust_hi = fs$ci[3, 2], h = fs$bws[1, 1]), 3)
#> jump_in_D robust_lo robust_hi         h
#>     0.625     0.595     0.652    18.496
```

8.2.2 Step 2: the reduced form (intention to treat)

```
rdplot(y = spp$Y, x = spp$X1, c = 0,
       x.label = "SISBEN distance to cutoff (positive = eligible)",
       y.label = "enrolled in college", title = "")
```

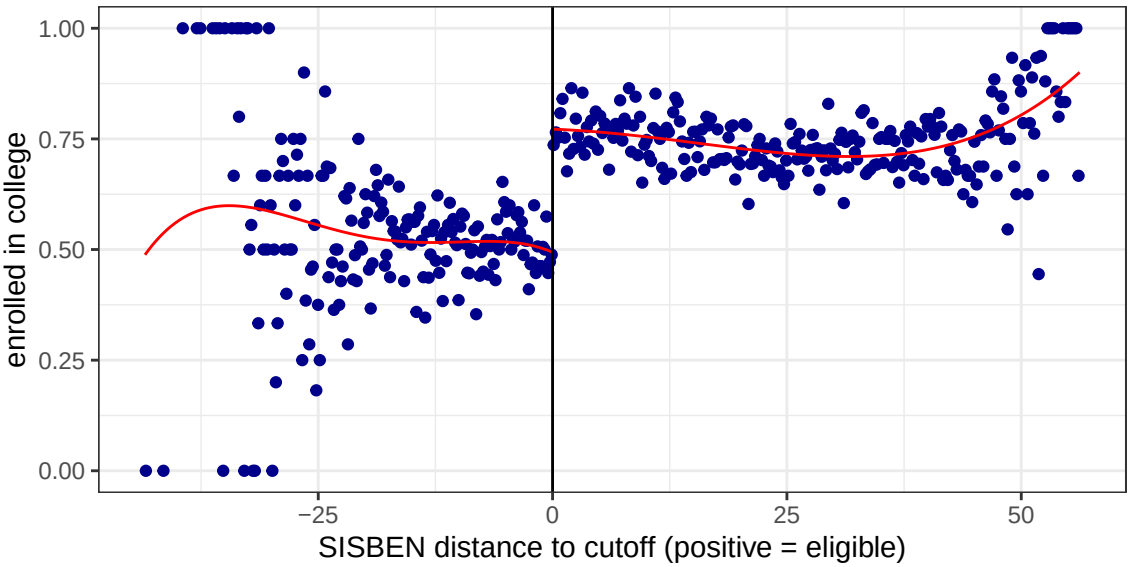


Figure 8: College enrollment against distance to the SISBEN cutoff: the effect of being eligible.

```
## reduced form as a sharp RD with Y as the outcome
rf <- rdrobust(y = spp$Y, x = spp$X1, c = 0)
round(c(jump_in_Y = rf$coef[1], robust_lo = rf$ci[3, 1],
        robust_hi = rf$ci[3, 2], h = rf$bws[1, 1]), 3)
#> jump_in_Y robust_lo robust_hi         h
#>     0.269     0.221     0.328     9.041
## dividing these two is NOT the fuzzy estimate: they use different bandwidths
round(rf$coef[1] / fs$coef[1], 3)
#> [1] 0.431
```

The reduced form is worth reporting on its own: being eligible raised enrollment by about 27 points. It needs no first-stage assumption.

8.2.3 Step 3: the fuzzy estimate

```
## one bandwidth, chosen for the ratio; both stages run at that bandwidth
fz <- rdrobust(y = spp$Y, x = spp$X1, c = 0, fuzzy = spp$D)
summary(fz)
#> Call: rdrobust
#>
#> Fuzzy RD estimates using local polynomial regression.
#>
#> Number of Obs.                23132
#> BW type                       mserd
#> Kernel                   Triangular
#> VCE method                       NN
#>
#>                                Left        Right
#> Number of Obs.                 7709        15423
#> Eff. Number of Obs.            3877         3908
#> Order est. (p)                    1            1
#> Order bias (q)                    2            2
#> BW est. (h)                   9.041        9.041
#> BW bias (b)                  14.402       14.402
#> rho (h/b)                     0.628        0.628
#> Unique Obs.                    3644         9274
#>
#> First-stage estimates.
#>
#> ====================================================================
#>                    Point    Robust Inference
#>                 Estimate         z     P>|z|      [ 95% C.I. ]
#> ====================================================================
#>      Rd Effect     0.619    29.918     0.000     [0.575 , 0.656]
#> ====================================================================
#>
#> Treatment effect estimates.
#>
#> ====================================================================
#>                    Point    Robust Inference
#>                 Estimate         z     P>|z|      [ 95% C.I. ]
#> --------------------------------------------------------------------
#>      RD Effect     0.435    11.023     0.000     [0.366 , 0.524]
#> ====================================================================
```

Two blocks in that output. The **first-stage** block is the jump in `D` at the fuzzy bandwidth (0.62,

not the 0.625 from Step 1, which used its own bandwidth). The **treatment-effect** block is the ratio. Read the robust row, as in the sharp case.

```
## estimate, s.e., p-value, 95% interval, three rows
round(cbind(estimate = fz$coef, se = fz$se, p = fz$pv, fz$ci), 3)
#>                Coeff Std. Err. P>|z| CI Lower CI Upper
#> Conventional   0.435     0.034     0    0.368    0.501
#> Bias-Corrected 0.445     0.034     0    0.378    0.512
#> Robust         0.445     0.040     0    0.366    0.524
## first stage at the fuzzy bandwidth, and that bandwidth
round(c(first_stage = fz$tau_T[1], h = fz$bws[1, 1]), 3)
#> first_stage           h
#>       0.619       9.041
## strength of the first stage: F = t^2 of the jump in D, as in Week 7
round((fz$tau_T[1] / fz$se_T[1])^2)
#> [1] 1288
```

An F above 1,000 is far from weak: the ratio divides by a large, precise number.

8.2.4 Step 4: the same number two ways, the ratio at one h and local 2SLS

```
library(ivreg)
h <- fz$bws[1, 1]
s <- spp |> filter(abs(X1) <= h) |>
  mutate(kw = 1 - abs(X1) / h)                  # triangular kernel

## stage 1: D on the instrument T, with a slope on each side
st1 <- lm(D ~ T + X1 + T:X1, data = s, weights = kw)
s$Dhat <- fitted(st1)
## stage 2: Y on predicted D, same slope controls
st2 <- lm(Y ~ Dhat + X1 + T:X1, data = s, weights = kw)
## the same in one call, with the 2SLS standard error
iv <- ivreg(Y ~ D + X1 + T:X1 | T + X1 + T:X1, data = s, weights = kw)
## Route 1 of the lecture: the reduced form at the SAME h, divided by stage 1
rf_h <- lm(Y ~ T + X1 + T:X1, data = s, weights = kw)

round(c(first_stage = coef(st1)[["T"]], reduced_form = coef(rf_h)[["T"]],
        ratio = coef(rf_h)[["T"]] / coef(st1)[["T"]],
        second_stage = coef(st2)[["Dhat"]],
        ivreg = coef(iv)[["D"]], rdrobust = fz$coef[1]), 4)
#>  first_stage reduced_form        ratio second_stage        ivreg
#>       0.6192       0.2690       0.4345       0.4345       0.4345
#>     rdrobust
#>       0.4345
```

At one bandwidth the ratio of the two jumps (Route 1) and 2SLS (Route 2) agree exactly, unlike the ratio in Step 2. The fuzzy RD *is* a kernel-weighted 2SLS inside the bandwidth. Use `rdrobust` for the interval: the hand-made second stage treats $\hat{D}$ as data, and neither `lm` nor `ivreg` corrects for the bias of the local line.

8.2.5 Step 5: checks

The sharp-RD checks apply to the running variable and to the reduced form.

```
## can families place themselves just below the SISBEN cutoff?
dens_spp <- rddensity(X = spp$X1, c = 0)
round(c(T_robust = dens_spp$test$t_jk, p_robust = dens_spp$test$p_jk), 3)
#> T_robust p_robust
#>   -0.799    0.424
```

```
## pre-determined covariates should not jump at the cutoff
cv_spp <- c("icfes_female", "icfes_age", "icfes_urm", "icfes_stratum",
            "icfes_famsize")
bal_spp <- sapply(cv_spp, function(z) {
  ok <- !is.na(spp[[z]])
  r <- rdrobust(y = spp[[z]][ok], x = spp$X1[ok], c = 0, bwselect = "cerrd")
  c(estimate = r$coef[2], robust_lo = r$ci[3, 1], robust_hi = r$ci[3, 2],
    p_robust = r$pv[3])
})
round(t(bal_spp), 3)
#>               estimate robust_lo robust_hi p_robust
#> icfes_female    -0.023    -0.092     0.046    0.517
#> icfes_age        0.146    -0.164     0.455    0.357
#> icfes_urm       -0.009    -0.035     0.017    0.494
#> icfes_stratum    0.032    -0.055     0.119    0.470
#> icfes_famsize   -0.120    -0.286     0.046    0.155
```

```
## fuzzy RD with covariates, on students with all five observed
sk <- spp[complete.cases(spp[, cv_spp]), ]
f0 <- rdrobust(y = sk$Y, x = sk$X1, c = 0, fuzzy = sk$D)
fzz <- rdrobust(y = sk$Y, x = sk$X1, c = 0, fuzzy = sk$D,
                covs = as.matrix(sk[, cv_spp]))
round(rbind(no_covariates = c(estimate = f0$coef[2], robust_se = f0$se[3],
                              f0$ci[3, ]),
            covariates = c(fzz$coef[2], fzz$se[3], fzz$ci[3, ])), 3)
#>               estimate robust_se CI Lower CI Upper
#> no_covariates    0.444     0.041    0.364    0.524
```

```
#> covariates       0.446     0.041    0.367    0.526
```

8.2.6 Reading the result

Receiving SPP raised immediate college enrollment by about 44 percentage points for **compliers at the cutoff**: students just poor enough to qualify who took up the subsidy. With one-sided non-compliance these are all the recipients near the cutoff, so the estimand is the effect on the treated at the cutoff. It says nothing directly about students far below the threshold, or about students below the exam cutoff, who were never in the sample.

9 Exercises

1. **Report an RD properly.** Write a short results paragraph for the Senate application containing: the robust estimate and interval, the bandwidth and effective n, the density test, one placebo, and one sentence stating the estimand. Under 150 words.
2. **Kernel choice.** Re-estimate with `kernel = "uniform"` and `kernel = "epanechnikov"`. How much does the answer move relative to its standard error? Which choice matters more, kernel or bandwidth?
3. **Local polynomial order.** Add `p = 3` to the comparison in `rd-order`. Note both the estimate and the selected bandwidth. Why does the optimal bandwidth widen with p, and why does the interval still widen?
4. **Covariates.** Repeat `covs` with one covariate at a time. Which one moves the estimate most, and does it match the covariate with the largest jump in `balance`?
5. **Detecting your own manipulation.** In `manip-sim`, vary the strength of the push (the `u > 0.5` threshold). At what point does the density test start to detect it? What does that tell you about the test's power?
6. **A real design.** Find an RD in the sociological literature. Identify the running variable, the cutoff, and the estimand in one sentence each. Then list which items of the lecture's "Reading and Reporting an RD" checklist the paper reports.
7. **Fuzzy RD on SPP.** Re-run `spp-fuzzy` with `bwselect = "cerrd"` and compare the bandwidth, estimate, and robust interval. Then name one way that being *eligible* could change enrollment other than through receiving the subsidy. Which assumption would that violate, and in which direction would it bias the fuzzy estimate?

10 Session information

```
sessionInfo()
#> R version 4.5.2 (2025-10-31)
```

```
#> Platform: aarch64-apple-darwin20
#> Running under: macOS Tahoe 26.6.2
#>
#> Matrix products: default
#> BLAS:
↪  /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Ve
↪
#> LAPACK:
↪  /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
↪  LAPACK version 3.12.1
#>
#> locale:
#> [1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
#>
#> time zone: America/New_York
#> tzcode source: internal
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#> [1] ivreg_0.6-8    rddensity_3.0  rdrobust_4.0.0 ggplot2_4.0.2
#> [5] dplyr_1.2.0
#>
#> loaded via a namespace (and not attached):
#>  [1] gtable_0.3.6       jsonlite_2.0.0     compiler_4.5.2
#>  [4] tidyselect_1.2.1   tinytex_0.58       scales_1.4.0
#>  [7] yaml_2.3.12        fastmap_1.2.0      lattice_0.22-7
#> [10] R6_2.6.1           labeling_0.4.3     generics_0.1.4
#> [13] lmtest_0.9-40      Formula_1.2-5      knitr_1.51
#> [16] MASS_7.3-65        tibble_3.3.1       car_3.1-5
#> [19] pillar_1.11.1      RColorBrewer_1.1-3 rlang_1.3.0
#> [22] xfun_0.56          S7_0.2.1           otel_0.2.0
#> [25] cli_3.6.6          withr_3.0.2        magrittr_2.0.4
#> [28] digest_0.6.39      grid_4.5.2         lifecycle_1.0.5
#> [31] lpdensity_3.0.1    vctrs_0.7.1        evaluate_1.0.5
#> [34] glue_1.8.0         farver_2.1.2       zoo_1.8-15
#> [37] abind_1.4-8        carData_3.0-6      rmarkdown_2.30
#> [40] tools_4.5.2        pkgconfig_2.0.3    htmltools_0.5.9
```